Sparsity and convergence analysis of Generalized Conditional Gradient Methods

Silvio Fanzon

Based on joint works with

Kristian Bredies, Marcello Carioni, Francisco Romero, Daniel Walter

13 September 2022

Frank-Wolfe and Generalized Conditional Gradient methods

Classical Frank-Wolfe-type algorithms aim at solving

$$\min_{x \in C} \; F(x)$$

where C is a convex set in $\mathbb{R}^N$ and $F \colon \mathbb{R}^N \to \mathbb{R}$ is regular convex function

Frank-Wolfe and Generalized Conditional Gradient methods

Classical Frank-Wolfe-type algorithms aim at solving

$$\min_{x \in C} \; F(x)$$

where C is a convex set in $\mathbb{R}^N$ and $F \colon \mathbb{R}^N \to \mathbb{R}$ is regular convex function

Algorithm: Given an iterate x^n, compute x^{n+1} in two steps:

- **Insertion step**: Solve the linearized problem around x^n as

$$\widehat{x} \in \operatorname{argmin}_{x \in C} \, \langle \nabla F(x^n), x \rangle$$

Frank-Wolfe and Generalized Conditional Gradient methods

Classical Frank-Wolfe-type algorithms aim at solving

$$\min_{x \in C} \; F(x)$$

where C is a convex set in $\mathbb{R}^N$ and $F\colon \mathbb{R}^N \to \mathbb{R}$ is regular convex function

Algorithm: Given an iterate x^n, compute x^{n+1} in two steps:

- **Insertion step**: Solve the linearized problem around x^n as

$$\widehat{x} \in \operatorname{argmin}_{x \in C} \, \langle \nabla F(x^n), x \rangle$$

- **Line search step**: Obtain x^{n+1} by interpolating

$$x^{n+1} = x^n + s^*(\widehat{x} - x^n)$$

  for a suitably chosen step-size s^*

Known facts about Frank-Wolfe

- The convergence rate is typically sublinear and it can be improved to linear under strong convexity assumptions on F and other interpolation steps [1] [2].
- The algorithm has been generalized to infinite dimensional spaces (Generalized Conditional Gradient methods) [3]
- Classical algorithms in infinite dimensional optimization have been shown to be particular instances of GCG [4] [5]

[1] Revisiting Frank-Wolfe: Projection-Free Sparse Convex Optimization Jaggi, M. (2013)
[2] On the Global Linear Convergence of Frank-Wolfe Optimization Variants Lacoste-Julien, S. and Jaggi, M. (2015)
[3] Approximate methods in optimization problems Demyanov, V. F. and Rubinov A. M. (1970)
[4] An iterative thresholding algorithm for linear inverse problems with a sparsity constraint Daubechies, I., Defrise, M. and De Mol, C. (2004)
[5] Iterated hard shrinkage for minimization problems with sparsity constraints Bredies, K. Lorenz, D. (2006)

GCG in the space of measures

Consider the BLASSO problem

$$\min_{\mu \in \mathcal{M}(\Omega)} G(\mu) := F(K\mu) + \|\mu\|_{\mathcal{M}(\Omega)}$$

with $\Omega \subset \mathbb{R}^d$ domain, $K : \mathcal{M}(\Omega) \to Y$ linear and weak*-to-strong continuous, Y Hilbert space, $F : Y \to [0, \infty)$ strictly convex and smooth

GCG in the space of measures

Consider the BLASSO problem

$$\min_{\mu \in \mathcal{M}(\Omega)} G(\mu) := F(K\mu) + \|\mu\|_{\mathcal{M}(\Omega)}$$

with $\Omega \subset \mathbb{R}^d$ domain, $K : \mathcal{M}(\Omega) \to Y$ linear and weak*-to-strong continuous, Y Hilbert space, $F : Y \to [0, \infty)$ strictly convex and smooth

Algorithm: Given an iterate μ^n one computes the next one μ^{n+1} in two steps:

- **Insertion step**: Solve the partially linearized problem around μ^n as

$$\widehat{\mu} \in \operatorname{argmax}_{\|\mu\|_{\mathcal{M}(\Omega)} \leq C} \langle -K_* \nabla F(K\mu^n), \mu \rangle$$

GCG in the space of measures

Consider the BLASSO problem

$$\min_{\mu \in \mathcal{M}(\Omega)} G(\mu) := F(K\mu) + \|\mu\|_{\mathcal{M}(\Omega)}$$

with $\Omega \subset \mathbb{R}^d$ domain, $K : \mathcal{M}(\Omega) \to Y$ linear and weak*-to-strong continuous, Y Hilbert space, $F : Y \to [0, \infty)$ strictly convex and smooth

Algorithm: Given an iterate μ^n one computes the next one μ^{n+1} in two steps:

- **Insertion step**: Solve the partially linearized problem around μ^n as

$$\widehat{\mu} \in \operatorname{argmax}_{\|\mu\|_{\mathcal{M}(\Omega)} \leq C} \langle -K_* \nabla F(K\mu^n), \mu \rangle$$

- **Line search step**: μ^{n+1} is obtained interpolating

$$\mu^{n+1} = \mu^n + s^*(\widehat{\mu} - \mu^n), \quad s^* \text{ suitable step-size}$$

GCG in the space of measures

Consider the BLASSO problem

$$\min_{\mu \in \mathcal{M}(\Omega)} G(\mu) := F(K\mu) + \|\mu\|_{\mathcal{M}(\Omega)}$$

with $\Omega \subset \mathbb{R}^d$ domain, $K : \mathcal{M}(\Omega) \to Y$ linear and weak*-to-strong continuous, Y Hilbert space, $F : Y \to [0, \infty)$ strictly convex and smooth

Algorithm: Given an iterate μ^n one computes the next one μ^{n+1} in two steps:

- **Insertion step**: Solve the partially linearized problem around μ^n as

$$\widehat{\mu} \in \operatorname{argmax}_{\|\mu\|_{\mathcal{M}(\Omega)} \leq C} \langle -K_* \nabla F(K\mu^n), \mu \rangle$$

- **Line search step**: μ^{n+1} is obtained interpolating

$$\mu^{n+1} = \mu^n + s^*(\widehat{\mu} - \mu^n), \quad s^* \text{ suitable step-size}$$

Inverse problems in the space of measures K. Bredies, H.K. Pikkarainen. ESAIM:COCV (2013)
The Alternating Descent Conditional Gradient Method for Sparse Inverse Problems N. Boyd, G. Schiebinger, B. Recht. *SIAM Journal on Optimization* (2017)

Key observation: iterate μ^n can be constructed as a combination of Dirac deltas

$$\mu^n = \sum_{i=1}^{k} c_i \, \delta_{x_i}$$

for suitable $c_i \in \mathbb{R}$ and $x_i \in \Omega$

Key observation: iterate μ^n can be constructed as a combination of Dirac deltas

$$\mu^n = \sum_{i=1}^{k} c_i \, \delta_{x_i}$$

for suitable $c_i \in \mathbb{R}$ and $x_i \in \Omega$

Why: Define the dual variable at n-th iteration as

$$p^n := -K_* \nabla F(K\mu^n) \in C(\Omega)$$

Key Lemma

A solution to

$$\max_{\|\mu\|_{\mathcal{M}(\Omega)} \leq C} \langle \mu, p^n \rangle$$

is given by $c\,\delta_{\hat{x}}$ for some $c \in \mathbb{R}$ and

$$\hat{x} \in \arg\max_{x \in \Omega} |p^n(x)|$$

The next iterate μ^{n+1} is obtained by

$$\mu^{n+1} = \mu^n + s^*(\widehat{\mu} - \mu^n)\,, \quad s^* \text{ suitable step-size}$$

where

$$\widehat{\mu} = c\,\delta_{\hat{x}}\,, \quad \hat{x} \in \arg\max_{x\in\Omega} |p^n(x)|$$

The next iterate μ^{n+1} is obtained by

$$\mu^{n+1} = \mu^n + s^*(\widehat{\mu} - \mu^n)\,, \quad s^* \text{ suitable step-size}$$

where

$$\widehat{\mu} = c\,\delta_{\hat{x}}\,, \quad \hat{x} \in \arg\max_{x\in\Omega} |p^n(x)|$$

Theorem (Bredies, Pikkarainen (2013))

μ^n converges weakly (up to subsequences) to a minimizer of G. Moreover the rate of convergence is sublinear, i.e.*

$$G(\mu^n) - \min_{\mu\in\mathcal{M}(\Omega)} G(\mu) \leq \frac{C}{n}$$

Remarks:

- The GCG method for BLASSO exploits sparsity of the problem (iterates are linear combinations of Dirac deltas)

Remarks:

- The GCG method for BLASSO exploits sparsity of the problem (iterates are linear combinations of Dirac deltas)
- GCG allows to design a grid-free algorithm that does not need an a priori discretization of the domain - we only need to find the max of $|p^n|$

Remarks:

- The GCG method for BLASSO exploits sparsity of the problem (iterates are linear combinations of Dirac deltas)
- GCG allows to design a grid-free algorithm that does not need an a priori discretization of the domain - we only need to find the max of $|p^n|$
- The Line Search Step can be be improved by optimizing the coefficients $(c_1, \dots, c_{k+1})$ of the full linear combination

$$\sum_{i=1}^{k} c_i \, \delta_{x_i} + c_{k+1} \, \delta_{\hat{x}}$$

with respect the energy G:

$$(c_i^*)_i \in \operatorname{argmin}_{c_i \in \mathbb{R}} \; G\left(\sum_{i=1}^{k} c_i \, \delta_{x_i} + c_{k+1} \, \delta_{\hat{x}}\right) \rightsquigarrow \mu^{n+1} = \sum_{i=1}^{k} c_i^* \, \delta_{x_i} + c_{k+1}^* \, \delta_{\hat{x}}$$

Remarks:

- The GCG method for BLASSO exploits sparsity of the problem (iterates are linear combinations of Dirac deltas)
- GCG allows to design a grid-free algorithm that does not need an a priori discretization of the domain - we only need to find the max of $|p^n|$
- The Line Search Step can be be improved by optimizing the coefficients $(c_1, \ldots, c_{k+1})$ of the full linear combination

$$\sum_{i=1}^{k} c_i \, \delta_{x_i} + c_{k+1} \, \delta_{\hat{x}}$$

with respect the energy G:

$$(c_i^*)_i \in \operatorname{argmin}_{c_i \in \mathbb{R}} \; G\left(\sum_{i=1}^{k} c_i \, \delta_{x_i} + c_{k+1} \, \delta_{\hat{x}}\right) \rightsquigarrow \mu^{n+1} = \sum_{i=1}^{k} c_i^* \, \delta_{x_i} + c_{k+1}^* \, \delta_{\hat{x}}$$

Remark: Coefficient optimization $\rightsquigarrow$ **linear** rate of convergence in experiments

Linear convergence in the space of measures

Question: Is it possible prove linear convergence?

[6] Linear convergence of generalized conditional gradient methods in the space of measures K. Pieper, D. Walter. ESAIM: COCV (2021)

[7] On the linear convergence rates of exchange and continuous methods for total variation minimization A. Flinth, F. De Gournay, P. Weiss. Math. Prog. (2021)

Linear convergence in the space of measures

Question: Is it possible prove linear convergence?

Define the dual variable of a minimizer $\bar{\mu}$ of G

$$\bar{p} = -K_* \nabla F(K\bar{\mu}) \in C(\Omega)$$

We make the following set of assumptions [6] [7]:

- Strong convexity of F, uniqueness and sparsity of the minimizer

(i) F is strongly convex
(ii) There exists $\{x_i\}_i \subset \Omega$ finite such that $\operatorname{argmax}_x |\bar{p}(x)| = \{x_i\}_i$
(iii) The set $\{K\delta_{x_i}\}_i \subset Y$ is linearly independent in Y

[6] Linear convergence of generalized conditional gradient methods in the space of measures K. Pieper, D. Walter. ESAIM: COCV (2021)

[7] On the linear convergence rates of exchange and continuous methods for total variation minimization A. Flinth, F. De Gournay, P. Weiss. Math. Prog. (2021)

Linear convergence in the space of measures

Question: Is it possible prove linear convergence?

Define the dual variable of a minimizer $\bar{\mu}$ of G

$$\bar{p} = -K_* \nabla F(K\bar{\mu}) \in C(\Omega)$$

We make the following set of assumptions [6] [7]:

- Strong convexity of F, uniqueness and sparsity of the minimizer

(i) F is strongly convex
(ii) There exists $\{x_i\}_i \subset \Omega$ finite such that $\operatorname{argmax}_x |\bar{p}(x)| = \{x_i\}_i$
(iii) The set $\{K\delta_{x_i}\}_i \subset Y$ is linearly independent in Y

(i) + (ii) + (iii) imply that the minimizer $\bar{\mu}$ of G is **unique** and **sparse**, i.e.,

$$\bar{\mu} = \sum_i c_i \, \delta_{x_i}$$

[6] Linear convergence of generalized conditional gradient methods in the space of measures K. Pieper, D. Walter. ESAIM: COCV (2021)

[7] On the linear convergence rates of exchange and continuous methods for total variation minimization A. Flinth, F. De Gournay, P. Weiss. Math. Prog. (2021)

- Quadratic growth of $\nabla^2 \bar{p}$ around u_i

$$-\mathrm{sign}(\bar{p}(x_i))\langle \xi, \nabla^2 \bar{p}(x_i)\xi \rangle \gtrapprox |\xi|^2 \qquad \text{for all } i$$

- Quadratic growth of $\nabla^2 \bar{p}$ around u_i

$$-\mathrm{sign}(\bar{p}(x_i))\langle \xi, \nabla^2 \bar{p}(x_i)\xi \rangle \gtrapprox |\xi|^2 \qquad \text{for all } i$$

- Lipschitz growth of K around δ_{x_i}

$$\|K(\delta_x - \delta_{x_i})\|_Y \lesssim |x - x_i| \qquad \text{for all } i$$

- Quadratic growth of $\nabla^2 \bar{p}$ around u_i

$$-\mathrm{sign}(\bar{p}(x_i))\langle \xi, \nabla^2 \bar{p}(x_i)\xi \rangle \gtrapprox |\xi|^2 \qquad \text{for all } i$$

- Lipschitz growth of K around δ_{x_i}

$$\|K(\delta_x - \delta_{x_i})\|_Y \lesssim |x - x_i| \qquad \text{for all } i$$

Theorem (Pieper, Walter (2020))

Under the previous assumptions the rate of convergence of μ^n is linear, i.e., there exists $C > 0$, $\zeta \in [1/2, 1)$ such that

$$G(\mu^n) - \min_{\mu \in \mathcal{M}(\Omega)} G(\mu) \leq C\zeta^n$$

GCG methods in infinite dimensions

Goal: generalize the previous algorithm and convergence results to Banach spaces

GCG methods in infinite dimensions

Goal: generalize the previous algorithm and convergence results to Banach spaces

We consider minimization problems of the form

$$\min_{u \in X} G(u) := F(Au) + R(u)$$

where we assume:

- X separable Banach space with predual X_*
- $F : Y \to \mathbb{R}$ data fidelity term, Y Hilbert space
- $K : X \to Y$ linear continuous measurement operator
- $K_* : Y \to X_*$ pre-adjoint of K
- $R : X \to [0, \infty]$ **convex**, **1-homogeneous** and **coercive**

Linear convergence of accelerated generalized conditional gradient methods
K. Bredies, M. Carioni, S. Fanzon, D. Walter. Preprint (2021)

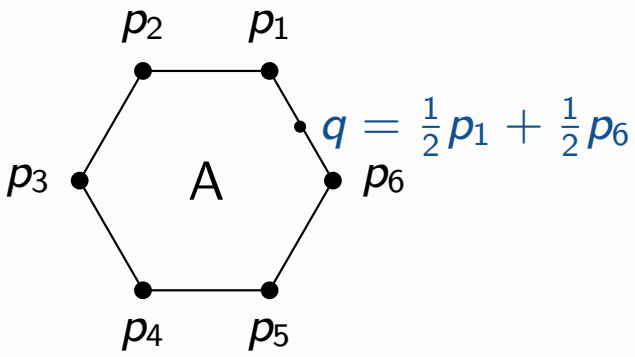
Question: How does **sparsity** enter the algorithm design?

Question: How does **sparsity** enter the algorithm design?

Extremal Points: Given a set B we say that $u \in B$ is an **extremal point** if

$$u = \lambda u_1 + (1-\lambda)u_2 \quad \text{for} \quad \lambda \in (0,1) \quad \Rightarrow \quad u = u_1 = u_2$$

The set of extremal points of B is denoted by $Ext(B)$

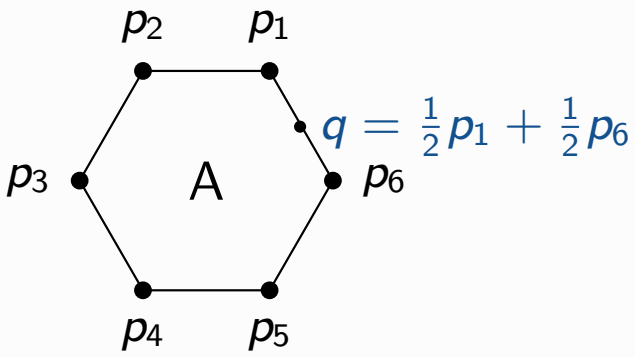


Question: How does **sparsity** enter the algorithm design?

Extremal Points: Given a set B we say that $u \in B$ is an **extremal point** if

$$u = \lambda u_1 + (1-\lambda)u_2 \quad \text{for} \quad \lambda \in (0,1) \quad \Rightarrow \quad u = u_1 = u_2$$

The set of extremal points of B is denoted by $Ext(B)$



Key Lemma

Define unit ball of the regularizer $B := \{u \in X : \ R(u) \leq 1\}$. Let $p \in X_*$. Then

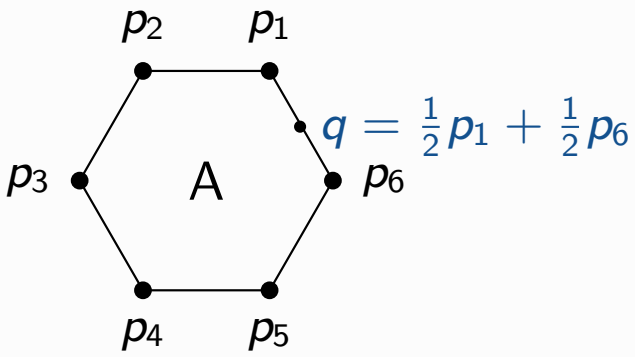
$$c\bar{v} \in \arg\max_{u \in B} \langle u, p \rangle, \quad c \in \mathbb{R}, \ \bar{v} \in \mathrm{Ext}(B)$$

Question: How does **sparsity** enter the algorithm design?

Extremal Points: Given a set B we say that $u \in B$ is an **extremal point** if

$$u = \lambda u_1 + (1-\lambda)u_2 \quad \text{for} \quad \lambda \in (0,1) \quad \Rightarrow \quad u = u_1 = u_2$$

The set of extremal points of B is denoted by $Ext(B)$



Key Lemma

Define unit ball of the regularizer $B := \{u \in X : \ R(u) \leq 1\}$. Let $p \in X_*$. Then

$$c\bar{v} \in \arg\max_{u \in B} \langle u, p \rangle, \quad c \in \mathbb{R}, \ \bar{v} \in \mathrm{Ext}(B)$$

In the insertion step we add an **extremal point** of B $\rightsquigarrow$ **Sparse** iterates

Accelerated Generalized Conditional Gradient algorithm

Assume the n-th iterate is sparse

$$u^n = \sum_{i=1}^{k} c_i u_i \,, \quad c_i \in \mathbb{R}, \quad u_i \in Ext(B)$$

Accelerated Generalized Conditional Gradient algorithm

Assume the n-th iterate is sparse

$$u^n = \sum_{i=1}^{k} c_i u_i \,, \quad c_i \in \mathbb{R}, \quad u_i \in Ext(B)$$

We compute u^{n+1} in the following way:

- **Insertion step:** Set $p^n = -K_* \nabla F(Ku^n)$ and $u_{k+1} := \widehat{u}$ where

$$\widehat{u} \in \operatorname{argmax}_{u \in B} \langle p^n, u \rangle \,, \quad \widehat{u} \in Ext(B)$$

Accelerated Generalized Conditional Gradient algorithm

Assume the n-th iterate is sparse

$$u^n = \sum_{i=1}^{k} c_i u_i \,, \quad c_i \in \mathbb{R}, \quad u_i \in Ext(B)$$

We compute u^{n+1} in the following way:

- **Insertion step:** Set $p^n = -K_* \nabla F(Ku^n)$ and $u_{k+1} := \widehat{u}$ where

$$\widehat{u} \in \operatorname{argmax}_{u \in B} \langle p^n, u \rangle \,, \quad \widehat{u} \in Ext(B)$$

- **Coefficients optimization step:** solve the finite-dimensional problem

$$(c_1^*, \ldots, c_{k+1}^*) \in \operatorname{argmin}_{c_i \in \mathbb{R}_+} \left[F\left(\sum_{i=1}^{k+1} c_i K u_i \right) + \sum_{i=1}^{k+1} c_i \right]$$

Accelerated Generalized Conditional Gradient algorithm

Assume the n-th iterate is sparse

$$u^n = \sum_{i=1}^{k} c_i u_i \,, \quad c_i \in \mathbb{R}, \quad u_i \in Ext(B)$$

We compute u^{n+1} in the following way:

- **Insertion step:** Set $p^n = -K_* \nabla F(Ku^n)$ and $u_{k+1} := \widehat{u}$ where

$$\widehat{u} \in \operatorname{argmax}_{u \in B} \langle p^n, u \rangle \,, \quad \widehat{u} \in Ext(B)$$

- **Coefficients optimization step:** solve the finite-dimensional problem

$$(c_1^*, \ldots, c_{k+1}^*) \in \operatorname{argmin}_{c_i \in \mathbb{R}_+} \left[F\left(\sum_{i=1}^{k+1} c_i K u_i \right) + \sum_{i=1}^{k+1} c_i \right]$$

Next iterate is

$$u^{n+1} := \sum_{i=1}^{k+1} c_i^* u_i$$

Worst-case convergence rate

Theorem (Bredies, Carioni, Fanzon, Walter (2021))

The iterate u^n weakly converges (up to subsequences) to a minimizer of G at a sublinear rate, i.e.*

$$G(u^n) - \min_u G(u) \leq \frac{C}{n}$$

[8] Sparsity of solutions for variational inverse problems with finite dimensional data
K. Bredies, M. Carioni. Calc. Var. PDE (2020)

Worst-case convergence rate

Theorem (Bredies, Carioni, Fanzon, Walter (2021))

The iterate u^n weakly converges (up to subsequences) to a minimizer of G at a sublinear rate, i.e.*

$$G(u^n) - \min_u G(u) \leq \frac{C}{n}$$

Remarks

- We recover the GCG for the minimization of the total variation noticing that

$$Ext(\{\mu : \|\mu\|_{\mathcal{M}(\Omega)} \leq 1\}) = \{\pm\delta_x : x \in \Omega\}$$

[8] Sparsity of solutions for variational inverse problems with finite dimensional data
K. Bredies, M. Carioni. Calc. Var. PDE (2020)

Worst-case convergence rate

Theorem (Bredies, Carioni, Fanzon, Walter (2021))

The iterate u^n weakly converges (up to subsequences) to a minimizer of G at a sublinear rate, i.e.*

$$G(u^n) - \min_u G(u) \leq \frac{C}{n}$$

Remarks

- We recover the GCG for the minimization of the total variation noticing that

$$Ext(\{\mu : \|\mu\|_{\mathcal{M}(\Omega)} \leq 1\}) = \{\pm\delta_x : x \in \Omega\}$$

- Wide class of applications: dynamic inverse problems, group sparsity regularizers, PDE constrained inverse problems, learning problems

[8] Sparsity of solutions for variational inverse problems with finite dimensional data
K. Bredies, M. Carioni. Calc. Var. PDE (2020)

Worst-case convergence rate

Theorem (Bredies, Carioni, Fanzon, Walter (2021))

The iterate u^n weakly converges (up to subsequences) to a minimizer of G at a sublinear rate, i.e.*

$$G(u^n) - \min_u G(u) \leq \frac{C}{n}$$

Remarks

- We recover the GCG for the minimization of the total variation noticing that

$$Ext(\{\mu : \|\mu\|_{\mathcal{M}(\Omega)} \leq 1\}) = \{\pm\delta_x : x \in \Omega\}$$

- Wide class of applications: dynamic inverse problems, group sparsity regularizers, PDE constrained inverse problems, learning problems
- Representer theorems show that extremal points of the regularizer are the natural atoms constituting sparse solutions [8]

[8] Sparsity of solutions for variational inverse problems with finite dimensional data
K. Bredies, M. Carioni. Calc. Var. PDE (2020)

Linear convergence

Goal: Find suitable assumptions under which linear convergence holds

Linear convergence

Goal: Find suitable assumptions under which linear convergence holds

Let $\bar{u}$ be a minimizer of G. Define the associated dual variable

$$\bar{p} := -K_* \nabla F(K\bar{u}) \in X_*$$

Assumptions:

- Strong convexity of F, uniqueness and sparsity of the minimizer

(i) F is strongly convex

(ii) There exists $\{u_i\}_i \subset Ext(B)$ s.t. $\operatorname{argmax}_{v \in \overline{Ext(B)}^*} \langle \bar{p}, v \rangle = \{u_i\}_i$

(iii) The set $\{Ku_i\}_i \subset Y$ is linearly independent in Y

Linear convergence

Goal: Find suitable assumptions under which linear convergence holds

Let $\bar{u}$ be a minimizer of G. Define the associated dual variable

$$\bar{p} := -K_* \nabla F(K\bar{u}) \in X_*$$

Assumptions:

- Strong convexity of F, uniqueness and sparsity of the minimizer

(i) F is strongly convex

(ii) There exists $\{u_i\}_i \subset Ext(B)$ s.t. $\operatorname{argmax}_{v \in \overline{Ext(B)}^*} \langle \bar{p}, v \rangle = \{u_i\}_i$

(iii) The set $\{Ku_i\}_i \subset Y$ is linearly independent in Y

(i) + (ii) + (iii) imply that the minimizer $\bar{u} \in X$ is unique and sparse, with

$$\bar{u} = \sum_i c_i u_i$$

There exists a function $g : Ext(B) \times Ext(B) \to [0, \infty)$ such that

- Quadratic growth of $\bar{p}$ around u_i

$$1 - \langle \bar{p}, u \rangle \gtrapprox g(u, u_i)^2 \quad \text{for every } i$$

There exists a function $g : Ext(B) \times Ext(B) \to [0, \infty)$ such that

- Quadratic growth of $\bar{p}$ around u_i

$$1 - \langle \bar{p}, u \rangle \gtrapprox g(u, u_i)^2 \quad \text{for every } i$$

- Lipschitz growth of K around u_i

$$\|K(u - u_i)\|_Y \lesssim g(u, u_i) \quad \text{for every } i$$

There exists a function $g : Ext(B) \times Ext(B) \to [0, \infty)$ such that

- Quadratic growth of $\bar{p}$ around u_i

$$1 - \langle \bar{p}, u \rangle \gtrapprox g(u, u_i)^2 \quad \text{for every } i$$

- Lipschitz growth of K around u_i

$$\|K(u - u_i)\|_Y \lesssim g(u, u_i) \quad \text{for every } i$$

Theorem (Bredies, Carioni, Fanzon, Walter (2021))

Under the previous assumptions the rate of convergence of u^n is linear, i.e. there exists $C > 0$, $\zeta \in [1/2, 1)$ s.t.

$$G(u^n) - \min_u G(u) \leq C\zeta^n$$

Lifting to the space of measures

Proof Strategy: **lift** the problem (and algorithm) to the space of measures on extremal points and prove convergence in the lifted space

Lifting to the space of measures

Proof Strategy: **lift** the problem (and algorithm) to the space of measures on extremal points and prove convergence in the lifted space

Denote $\mathcal{B} := \overline{Ext(B)}^*$ endowed with the metric that metrizes the weak* convergence. We consider positive measures on $\mathcal{B}$, i.e.

$$\mathcal{M}^+(\mathcal{B})$$

Definition

We say that a measure $\mu \in \mathcal{M}^+(\mathcal{B})$ represents $u \in X$ if

$$\langle p, u \rangle = \int_{\mathcal{B}} \langle p, v \rangle \, d\mu(v) \quad \forall p \in X_*$$

Alternatively, u is said to be the weak barycenter of μ

Lifting to the space of measures

Proof Strategy: **lift** the problem (and algorithm) to the space of measures on extremal points and prove convergence in the lifted space

Denote $\mathcal{B} := \overline{Ext(B)}^*$ endowed with the metric that metrizes the weak* convergence. We consider positive measures on $\mathcal{B}$, i.e.

$$\mathcal{M}^+(\mathcal{B})$$

Definition

We say that a measure $\mu \in \mathcal{M}^+(\mathcal{B})$ represents $u \in X$ if

$$\langle p, u \rangle = \int_{\mathcal{B}} \langle p, v \rangle \, d\mu(v) \quad \forall p \in X_*$$

Alternatively, u is said to be the weak barycenter of μ

Example: the measure δ_u represents u for all $u \in \mathcal{B}$

As a consequence of Choquet's Theorem we have:

Proposition (Bredies, Carioni, Fanzon, Walter)

- *For every* $\mu \in \mathcal{M}^+(\mathcal{B})$ *we have*

$$R(u) \leq \|\mu\|_{\mathcal{M}(\mathcal{B})}$$

  where $u \in dom(R)$ *is the weak barycenter of* μ
- *For every* $u \in dom(R)$ *there exists* $\mu \in \mathcal{M}^+(\mathcal{B})$ *concentrated on* $Ext(B)$ *that represents* u *and such that*

$$R(u) = \|\mu\|_{\mathcal{M}(\mathcal{B})}$$

As a consequence of Choquet's Theorem we have:

Proposition (Bredies, Carioni, Fanzon, Walter)

- *For every* $\mu \in \mathcal{M}^+(\mathcal{B})$ *we have*

$$R(u) \leq \|\mu\|_{\mathcal{M}(\mathcal{B})}$$

  where $u \in dom(R)$ *is the weak barycenter of* μ
- *For every* $u \in dom(R)$ *there exists* $\mu \in \mathcal{M}^+(\mathcal{B})$ *concentrated on* $Ext(B)$ *that represents* u *and such that*

$$R(u) = \|\mu\|_{\mathcal{M}(\mathcal{B})}$$

Lifted variational problem:

$$\min_{\mu \in \mathcal{M}^+(\mathcal{B})} \hat{G}(\mu) := F(\mathcal{K}\mu) + \|\mu\|_{\mathcal{M}(\mathcal{B})}$$

As a consequence of Choquet's Theorem we have:

Proposition (Bredies, Carioni, Fanzon, Walter)

- *For every* $\mu \in \mathcal{M}^+(\mathcal{B})$ *we have*

$$R(u) \leq \|\mu\|_{\mathcal{M}(\mathcal{B})}$$

  where $u \in dom(R)$ *is the weak barycenter of* μ
- *For every* $u \in dom(R)$ *there exists* $\mu \in \mathcal{M}^+(\mathcal{B})$ *concentrated on* $Ext(B)$ *that represents* u *and such that*

$$R(u) = \|\mu\|_{\mathcal{M}(\mathcal{B})}$$

Lifted variational problem:

$$\min_{\mu \in \mathcal{M}^+(\mathcal{B})} \hat{G}(\mu) := F(\mathcal{K}\mu) + \|\mu\|_{\mathcal{M}(\mathcal{B})}$$

$\mathcal{K} : \mathcal{M}^+(\mathcal{B}) \to Y$ is the lift of $K : X \to Y$ defined by the relation

$$(\mathcal{K}\mu, y)_Y = \int_{\mathcal{B}} (Kv, y)\, d\mu(v) \quad \forall y \in Y$$

The following problems are **equivalent**

$$\min_{u \in X} G(u) := F(Ku) + R(u) \qquad \text{(original problem)} \qquad \text{(OP)}$$

$$\min_{\mu \in \mathcal{M}^+(\mathcal{B})} \hat{G}(\mu) := F(\mathcal{K}\mu) + \|\mu\|_{\mathcal{M}(\mathcal{B})} \qquad \text{(lifted problem)} \qquad \text{(LP)}$$

The following problems are **equivalent**

$$\min_{u \in X} G(u) := F(Ku) + R(u) \qquad \text{(original problem)} \qquad \text{(OP)}$$

$$\min_{\mu \in \mathcal{M}^+(\mathcal{B})} \hat{G}(\mu) := F(\mathcal{K}\mu) + \|\mu\|_{\mathcal{M}(\mathcal{B})} \qquad \text{(lifted problem)} \qquad \text{(LP)}$$

Theorem (Bredies, Carioni, Fanzon, Walter)

- $\bar{u} \in X$ *solves* (OP) $\implies \exists\, \bar{\mu} \in \mathcal{M}^+(\mathcal{B})$ *that represents* $\bar{u}$ *and solves* (LP)
- $\bar{\mu} \in \mathcal{M}^+(\mathcal{B})$ *solves* (LP) $\implies$ *weak barycenter of* $\bar{\mu}$ *solves* (OP)

Idea: GCG algorithm for (LP) $\implies$ GCG algorithm for (OP) taking barycenters:

$$\mu = \sum_i c_i\, \delta_{u_i} \in \mathcal{M}^+(\mathcal{B}) \qquad \rightsquigarrow \qquad u = \sum_i c_i\, u_i \in X$$

$\Rightarrow$ Prove linear convergence for the Lifted Problem

Dynamic inverse problems

Inverse Problem: Given $f_t \in H_t$, find a curve $t \mapsto \rho_t \in \mathcal{M}(\Omega)$ s.t.

$$K_t^* \rho_t = f_t \quad \text{for a.e.} \quad t \in (0,1) \tag{P}$$

Dynamic inverse problems

Inverse Problem: Given $f_t \in H_t$, find a curve $t \mapsto \rho_t \in \mathcal{M}(\Omega)$ s.t.

$$K_t^* \rho_t = f_t \quad \text{for a.e.} \quad t \in (0,1) \tag{P}$$

- **Unknown:** curve of measures $t \mapsto \rho_t \in \mathcal{M}(\Omega)$, with $\Omega \subset \mathbb{R}^d$ bounded
- **Data**: curve $t \mapsto f_t \in H_t$ with $\{H_t\}_t$ family of Hilbert spaces
- **Measurements:** linear continuous operators $K_t^* : \mathcal{M}(\Omega) \to H_t$

Dynamic inverse problems

Inverse Problem: Given $f_t \in H_t$, find a curve $t \mapsto \rho_t \in \mathcal{M}(\Omega)$ s.t.

$$K_t^* \rho_t = f_t \quad \text{for a.e.} \quad t \in (0,1) \tag{P}$$

- **Unknown:** curve of measures $t \mapsto \rho_t \in \mathcal{M}(\Omega)$, with $\Omega \subset \mathbb{R}^d$ bounded
- **Data**: curve $t \mapsto f_t \in H_t$ with $\{H_t\}_t$ family of Hilbert spaces
- **Measurements:** linear continuous operators $K_t^* : \mathcal{M}(\Omega) \to H_t$

Assumptions: very weak time-regularity for $\{H_t\}_t$ and K_t^*

Dynamic inverse problems

Inverse Problem: Given $f_t \in H_t$, find a curve $t \mapsto \rho_t \in \mathcal{M}(\Omega)$ s.t.

$$K_t^* \rho_t = f_t \quad \text{for a.e.} \quad t \in (0,1) \tag{P}$$

- **Unknown:** curve of measures $t \mapsto \rho_t \in \mathcal{M}(\Omega)$, with $\Omega \subset \mathbb{R}^d$ bounded
- **Data**: curve $t \mapsto f_t \in H_t$ with $\{H_t\}_t$ family of Hilbert spaces
- **Measurements:** linear continuous operators $K_t^* : \mathcal{M}(\Omega) \to H_t$

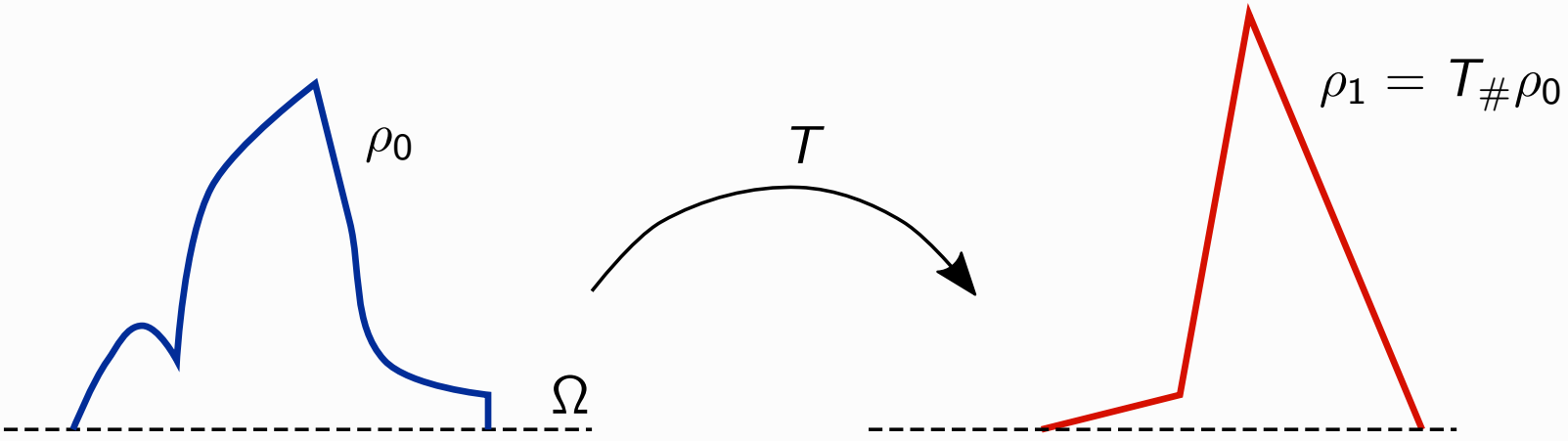
Assumptions: very weak time-regularity for $\{H_t\}_t$ and K_t^*

Goal: Regularize (P) with an Optimal Transport energy acting on the measure ρ_t
This will enforce time regularity of the reconstructions

An optimal transport approach for solving dynamic inverse problems in spaces of measures.
K. Bredies, S. Fanzon. ESAIM: M2AN (2020)

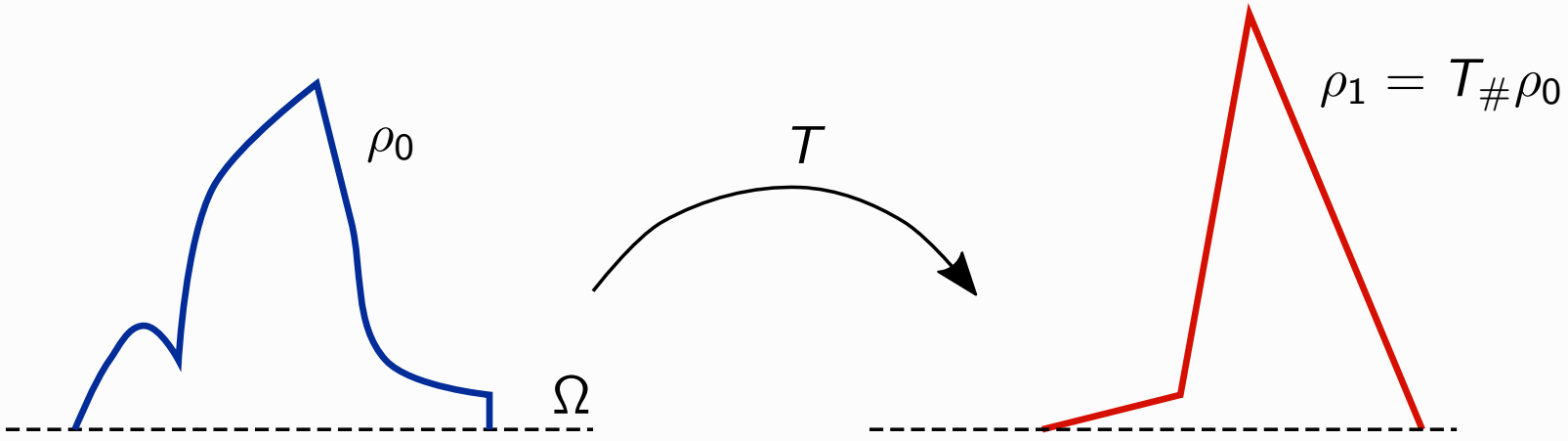
Optimal Transport - Static Formulation

$\Omega \subset \mathbb{R}^d$ bounded domain, $\rho_0, \rho_1 \in \mathcal{P}(\Omega)$, $T \colon \Omega \to \Omega$ measurable displacement



Optimal Transport - Static Formulation

$\Omega \subset \mathbb{R}^d$ bounded domain, $\rho_0, \rho_1 \in \mathcal{P}(\Omega)$, $T \colon \Omega \to \Omega$ measurable displacement

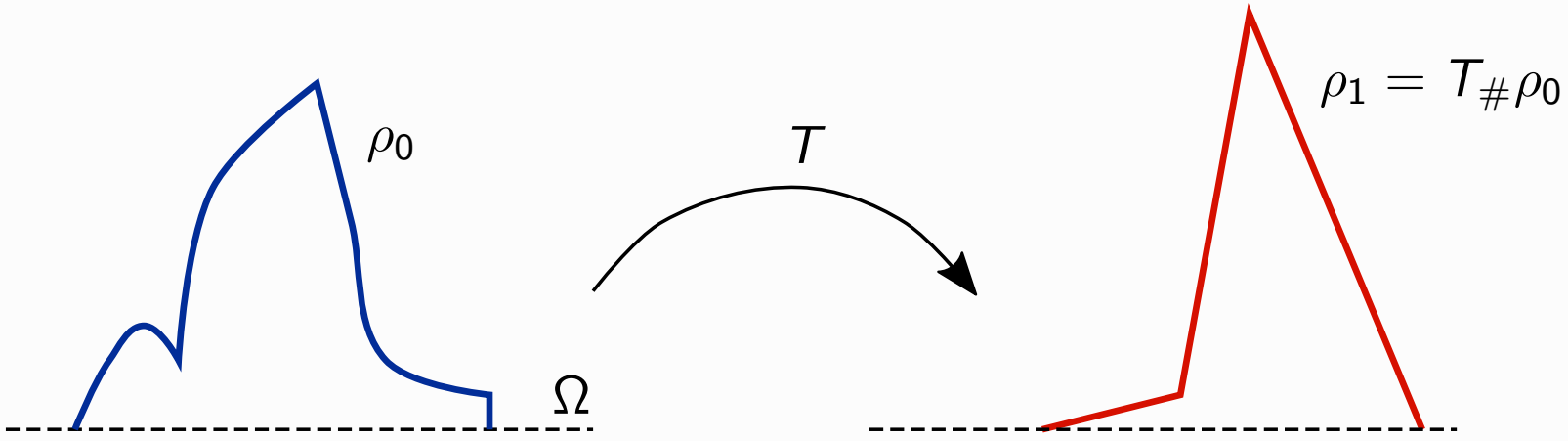


Goal: move ρ_0 to ρ_1 in the cheapest way, with cost of moving mass from x to y

$$c(x, y) := |x - y|^2$$

Optimal Transport - Static Formulation

$\Omega \subset \mathbb{R}^d$ bounded domain, $\rho_0, \rho_1 \in \mathcal{P}(\Omega)$, $T\colon \Omega \to \Omega$ measurable displacement



Goal: move ρ_0 to ρ_1 in the cheapest way, with cost of moving mass from x to y

$$c(x, y) := |x - y|^2$$

Optimal Transport: a transport plan $\hat{T}$ solving

$$\hat{T} \in \arg\min \left\{ \int_\Omega |T(x) - x|^2 \, d\rho_0(x) : \; T\colon \Omega \to \Omega, \; T_{\#}\rho_0 = \rho_1 \right\}$$

Optimal Transport - Dynamic Formulation

Idea: introduce a time variable $t \in [0, 1]$ and consider the density evolution

Optimal Transport - Dynamic Formulation

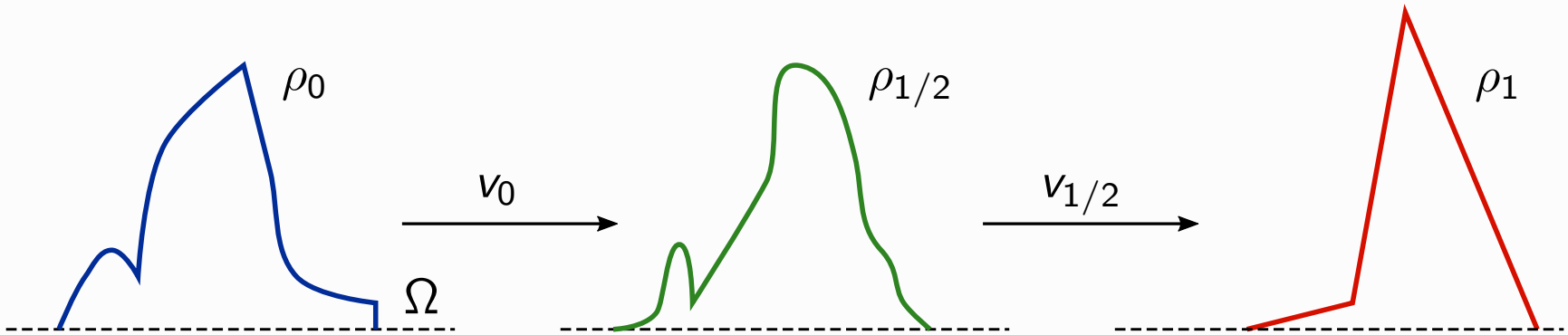
Idea: introduce a time variable $t \in [0,1]$ and consider the density evolution

- time dependent probability measures

$$t \mapsto \rho_t \in \mathcal{P}(\Omega) \quad \text{for} \quad t \in [0,1]$$

- ρ_t is advected by the velocity field

$$v_t(x) \colon [0,1] \times \Omega \to \mathbb{R}^d$$



Optimal Transport - Dynamic Formulation

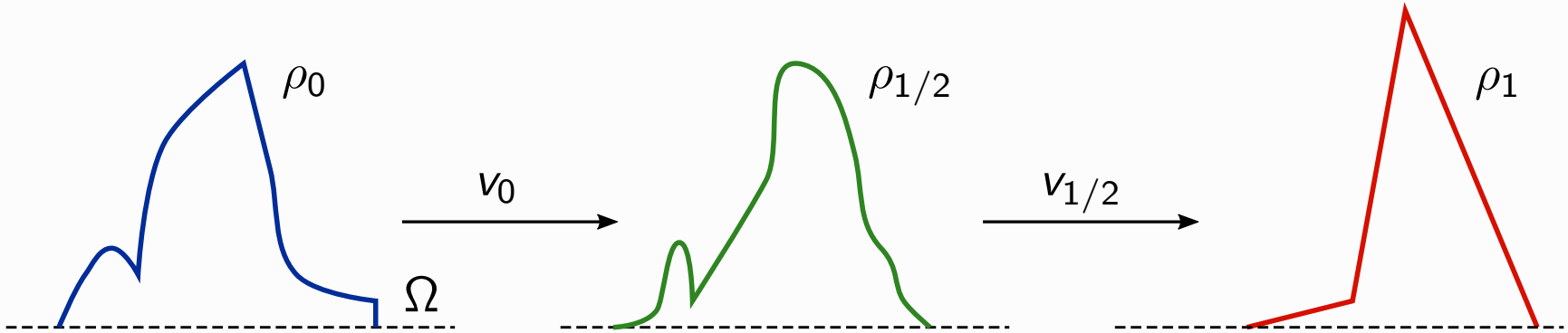
Idea: introduce a time variable $t \in [0,1]$ and consider the density evolution

- time dependent probability measures

$$t \mapsto \rho_t \in \mathcal{P}(\Omega) \quad \text{for} \quad t \in [0,1]$$

- ρ_t is advected by the velocity field

$$v_t(x) \colon [0,1] \times \Omega \to \mathbb{R}^d$$



Dynamic model: (ρ_t, v_t) solves the continuity equation with initial conditions

$$\begin{cases} \partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0 \\ \text{Initial data } \rho_0, \text{ final data } \rho_1 \end{cases} \qquad \text{(CE-IC)}$$

Connection and Advantages

Theorem (Benamou-Brenier '00)

$$\min_{\substack{(\rho_t, v_t) \\ \textit{solving } \text{(CE-IC)}}} \int_0^1 \int_\Omega |v_t(x)|^2 \rho_t(x) dx\, dt = \min_{\substack{T:\, \Omega \to \Omega \\ T_\# \rho_0 = \rho_1}} \int_\Omega |T(x) - x|^2 \rho_0(x)\, dx$$

Connection and Advantages

Theorem (Benamou-Brenier '00)

$$\min_{\substack{(\rho_t, v_t) \\ \text{solving (CE-IC)}}} \int_0^1 \int_\Omega |v_t(x)|^2 \rho_t(x) dx\, dt = \min_{\substack{T:\, \Omega \to \Omega \\ T_\# \rho_0 = \rho_1}} \int_\Omega |T(x) - x|^2 \rho_0(x)\, dx$$

Advantages of Dynamic Formulation:

1. By introducing the momentum $m_t := \rho_t v_t$ we have

$$\int_0^1 \int_\Omega |v_t(x)|^2 \rho_t(x)\, dx\, dt = \int_0^1 \int_\Omega \frac{|m_t(x)|^2}{\rho_t(x)}\, dx\, dt$$

which is **convex** in (ρ_t, m_t)

Connection and Advantages

Theorem (Benamou-Brenier '00)

$$\min_{\substack{(\rho_t, v_t) \\ \text{solving (CE-IC)}}} \int_0^1 \int_\Omega |v_t(x)|^2 \rho_t(x) dx\, dt = \min_{\substack{T:\, \Omega \to \Omega \\ T_\# \rho_0 = \rho_1}} \int_\Omega |T(x) - x|^2 \rho_0(x)\, dx$$

Advantages of Dynamic Formulation:

1. By introducing the momentum $m_t := \rho_t v_t$ we have

$$\int_0^1 \int_\Omega |v_t(x)|^2 \rho_t(x)\, dx\, dt = \int_0^1 \int_\Omega \frac{|m_t(x)|^2}{\rho_t(x)}\, dx\, dt$$

   which is **convex** in (ρ_t, m_t)

2. The continuity equation becomes **linear**

$$\partial_t \rho_t + \operatorname{div} m_t = 0$$

Connection and Advantages

Theorem (Benamou-Brenier '00)

$$\min_{\substack{(\rho_t, v_t) \\ \text{solving (CE-IC)}}} \int_0^1 \int_\Omega |v_t(x)|^2 \rho_t(x) dx\, dt = \min_{\substack{T:\, \Omega \to \Omega \\ T_\# \rho_0 = \rho_1}} \int_\Omega |T(x) - x|^2 \rho_0(x)\, dx$$

Advantages of Dynamic Formulation:

1. By introducing the momentum $m_t := \rho_t v_t$ we have

$$\int_0^1 \int_\Omega |v_t(x)|^2 \rho_t(x)\, dx\, dt = \int_0^1 \int_\Omega \frac{|m_t(x)|^2}{\rho_t(x)}\, dx\, dt$$

   which is **convex** in (ρ_t, m_t)

2. The continuity equation becomes **linear**

$$\partial_t \rho_t + \operatorname{div} m_t = 0$$

3. We know the full trajectory ρ_t and can recover the velocity field v_t from m_t

Optimal transport regularization

Time-space domain $X := (0,1) \times \Omega$, measures $\mathcal{M} := \mathcal{M}(X) \times \mathcal{M}(X; \mathbb{R}^d)$

We regularize $K_t^* \rho_t = f_t$ via minimization in $(\rho, m) \in \mathcal{M}$ of

$$G_{\alpha,\beta}(\rho, m) := \frac{1}{2} \int_0^1 \|K_t^* \rho_t - f_t\|_{H_t}^2 \, dt + J_{\alpha,\beta}(\rho, m)$$

where the optimal transport regularizer is

$$J_{\alpha,\beta}(\rho, m) := \frac{\alpha}{2} \underbrace{\int_0^1 \int_\Omega \left| \frac{dm}{d\rho} \right|^2 d\rho(t,x)}_{\text{Optimal Transport Regularizer}} + \beta \underbrace{\|\rho\|_{\mathcal{M}(X)}}_{\text{TV Regularizer}}$$

$$\text{s.t.} \quad \partial_t \rho_t + \operatorname{div} m_t = 0 \qquad \text{(Continuity Equation - No IC)}$$

Existence and Sparsity

Theorem (Bredies, Fanzon '20)

(With assumptions on f_t, K_t^*, H_t) The functional $G_{\alpha,\beta}$ **admits solution**

$$\rho = dt \otimes \rho_t\,, \;\; m = v\rho\,, \;\; v\colon X \to \mathbb{R}^d$$

with v measurable **velocity field** *and $t \mapsto \rho_t \in \mathcal{M}^+(\Omega)$* **narrowly continuous**.
Moreover we have **stability**

Existence and Sparsity

Theorem (Bredies, Fanzon '20)

(With assumptions on f_t, K_t^*, H_t) The functional $G_{\alpha,\beta}$ **admits solution**

$$\rho = dt \otimes \rho_t\,, \;\; m = v\rho\,, \;\; v\colon X \to \mathbb{R}^d$$

with v measurable **velocity field** *and $t \mapsto \rho_t \in \mathcal{M}^+(\Omega)$* **narrowly continuous**. *Moreover we have* **stability**

Definition: An **atom** in $\mathcal{M}$ is a pair (ρ_γ, m_γ) with $\gamma \in H^1([0,1];\Omega)$,

$$\rho_\gamma := a_\gamma\, dt \otimes \delta_{\gamma(t)}\,, \;\; m_\gamma := \dot\gamma(t)\,\rho_\gamma\,, \;\; a_\gamma := \left(\frac{\beta}{2}\int_0^1 |\dot\gamma(t)|^2\, dt + \alpha\right)^{-1}$$

Generalized Conditional Gradient Method

Theorem (Bredies, Carioni, Fanzon, Romero '20)

Consider the sublevel set $C_{\alpha,\beta} := \{J_{\alpha,\beta} \leq 1\}$. *The* **extremal points** *of* $C_{\alpha,\beta}$ *are*

$$\mathrm{Ext}(C_{\alpha,\beta}) = \{ \textit{ atoms } \} \cup (0,0)$$

Generalized Conditional Gradient Method

Theorem (Bredies, Carioni, Fanzon, Romero '20)

Consider the sublevel set $C_{\alpha,\beta} := \{J_{\alpha,\beta} \leq 1\}$. *The* **extremal points** *of* $C_{\alpha,\beta}$ *are*

$$\mathrm{Ext}(C_{\alpha,\beta}) = \{ \textit{ atoms } \} \cup (0,0)$$

Goal: Find numerical solutions to the minimization problem for $G_{\alpha,\beta}$ by GCG

Generalized Conditional Gradient Method

Theorem (Bredies, Carioni, Fanzon, Romero '20)

Consider the sublevel set $C_{\alpha,\beta} := \{J_{\alpha,\beta} \leq 1\}$. *The* **extremal points** *of* $C_{\alpha,\beta}$ *are*

$$\mathrm{Ext}(C_{\alpha,\beta}) = \{ \text{ atoms } \} \cup (0,0)$$

Goal: Find numerical solutions to the minimization problem for $G_{\alpha,\beta}$ by GCG

Key Step: Find a descent direction around $(\tilde{\rho}, \tilde{m})$ by solving

$$\min_{(\rho,m)\in C_{\alpha,\beta}} \; -\int_0^1 \langle \rho_t, w_t \rangle \, dt \,, \quad w_t := -K_t(K_t^* \tilde{\rho}_t - f_t) \in C(\Omega) \tag{D}$$

Generalized Conditional Gradient Method

Theorem (Bredies, Carioni, Fanzon, Romero '20)

Consider the sublevel set $C_{\alpha,\beta} := \{J_{\alpha,\beta} \leq 1\}$. The **extremal points** *of $C_{\alpha,\beta}$ are*

$$\mathrm{Ext}(C_{\alpha,\beta}) = \{ \text{ atoms } \} \cup (0,0)$$

Goal: Find numerical solutions to the minimization problem for $G_{\alpha,\beta}$ by GCG

Key Step: Find a descent direction around $(\tilde{\rho}, \tilde{m})$ by solving

$$\min_{(\rho,m)\in C_{\alpha,\beta}} \; -\int_0^1 \langle \rho_t, w_t \rangle \, dt \,, \quad w_t := -K_t(K_t^* \tilde{\rho}_t - f_t) \in C(\Omega) \qquad \text{(D)}$$

Theorem (SF, Bredies, Carioni, Romero '20)

Problem (D) admits a solution which is either an **atom** *or $(0,0)$.*

Therefore (D) can be casted in $H^1([0,1];\Omega)$, and is hence numerically feasible

Numerical Algorithm

Let $t \mapsto f_t$ be given data. Initialize $\rho^0 := 0$

1. **Insertion:** Given $\rho^n = \sum_{i=1}^N c_i \, \rho_{\gamma_i}$, set $w_t^n := -K_t(K_t^* \rho_t^n - f_t)$ and find

$$\gamma^* \in \arg\min_{\gamma \in H^1} -a_\gamma \int_0^1 w_t(\gamma(t))\, dt\,, \qquad \rho^{n+1/2} := \rho^n + c_{N+1}\, \rho_{\gamma^*}$$

2. **Coefficients Optimization:** Solve the quadratic problem

$$(c_j^*)_j \in \arg\min_{c_j \geq 0} \; G_{\alpha,\beta}(\rho^{n+1/2}, m^{n+1/2})\,, \qquad \rho^{n+1} := \sum_{i=1}^{N+1} c_i^* \, \rho_{\gamma_i}$$

Theorem (Bredies, Carioni, Fanzon, Romero '20)

The sequence (ρ^n, m^n) in Algorithm converges weak to a minimizer of $G_{\alpha,\beta}$*

The convergence rate is of order $O(1/n)$

Application: Undersampled Fourier Measurements

- $\Omega := [0,1]^2$ image frame, $t \mapsto \sigma_t \in \mathcal{M}^+(\mathbb{R}^2)$ frequencies sampling measure
- Fourier transform $\mathfrak{F}\colon \mathcal{M}(\Omega) \to C^\infty(\mathbb{R}^2;\mathbb{C})$
- $H_t := L^2_{\sigma_t}(\mathbb{R}^2;\mathbb{C})$ and $K_t^*\colon \mathcal{M}(\Omega) \to H_t$ defined by $K_t^*\rho := \mathfrak{F}\rho$

Note. K_t^* corresponds to the Fourier transform undersampled according to σ_t

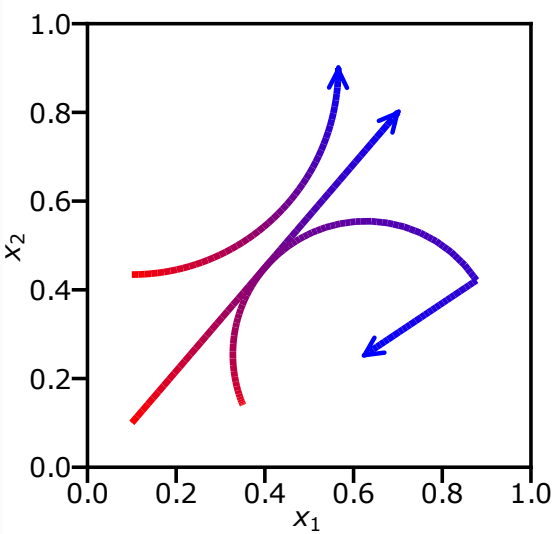
Time-discrete sampling: Fix $T+1$ times samples, $t_i := i/T$ for $i = 0, \dots, T$

- At each time t_i sample $n_i \in \mathbb{N}$ frequencies $\{S_{i,1}, \dots, S_{i,n_i}\} \subset \mathbb{R}^2$
- Define $t \mapsto \sigma_t$ so that $\sigma_{t_i} = \sum_{k=1}^{n_i} \delta_{S_{i,k}}$. In this case $H_{t_i} = \mathbb{C}^{n_i}$ and

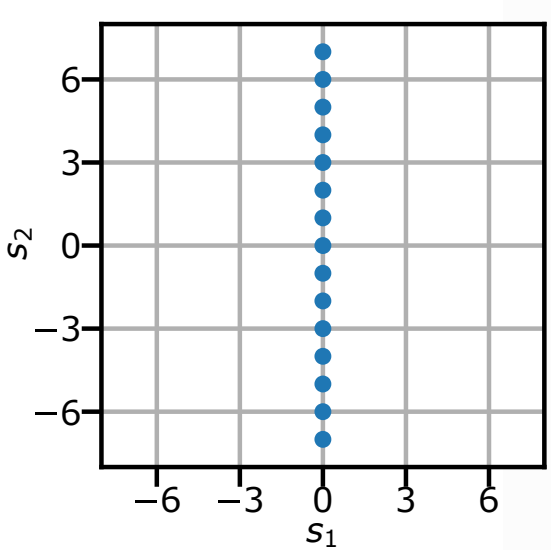
$$K_{t_i}^*\rho = \left(\int_{\mathbb{R}^2} \exp(-2\pi \mathbf{i} x \cdot S_{i,k})\, d\rho(x) \right)_{k=1}^{n_i} \in \mathbb{C}^{n_i}$$

Experiment: Dynamic Spikes Tracking

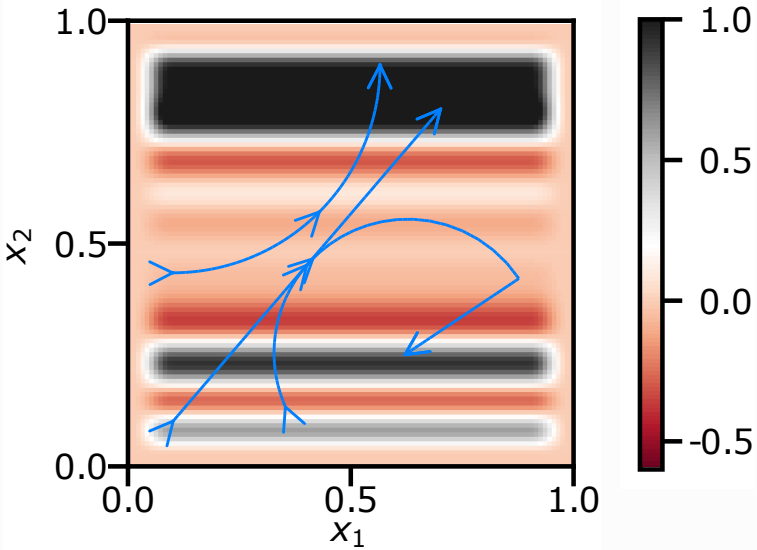
- $T = 50$, $n_i = 15$ freq. sampled on lines L_i through the origin with angle $\frac{i\pi}{4}$
- Ground Truth: $\tilde{\rho}_t = \delta_{\gamma_1(t)} + \delta_{\gamma_2(t)} + \delta_{\gamma_3(t)}$ as depicted (color=position in time)
- Synthetic Data: $f_{t_i} := K^*_{t_i}\tilde{\rho}_{t_i} + 60\%$ Gaussian Noise
- Data Visualization: By plotting the initial dual variable $w^0_{t_i} := K_{t_i} f_{t_i} \in C(\Omega)$



Ground Truth



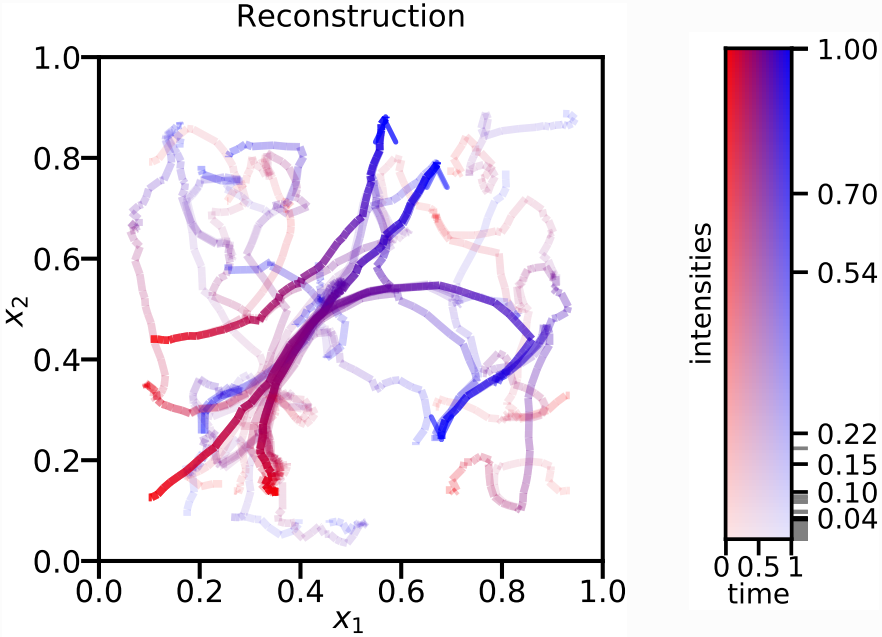
Sampled Freq. $t = 1$



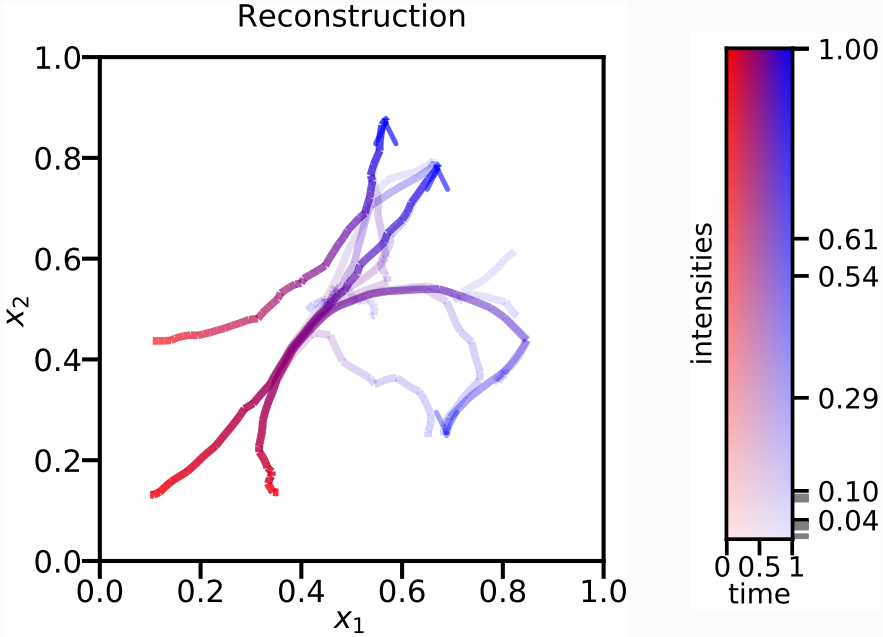
$w^0_t(x)$, 60% noise

Reconstructions

$\alpha = \beta = 0.1$



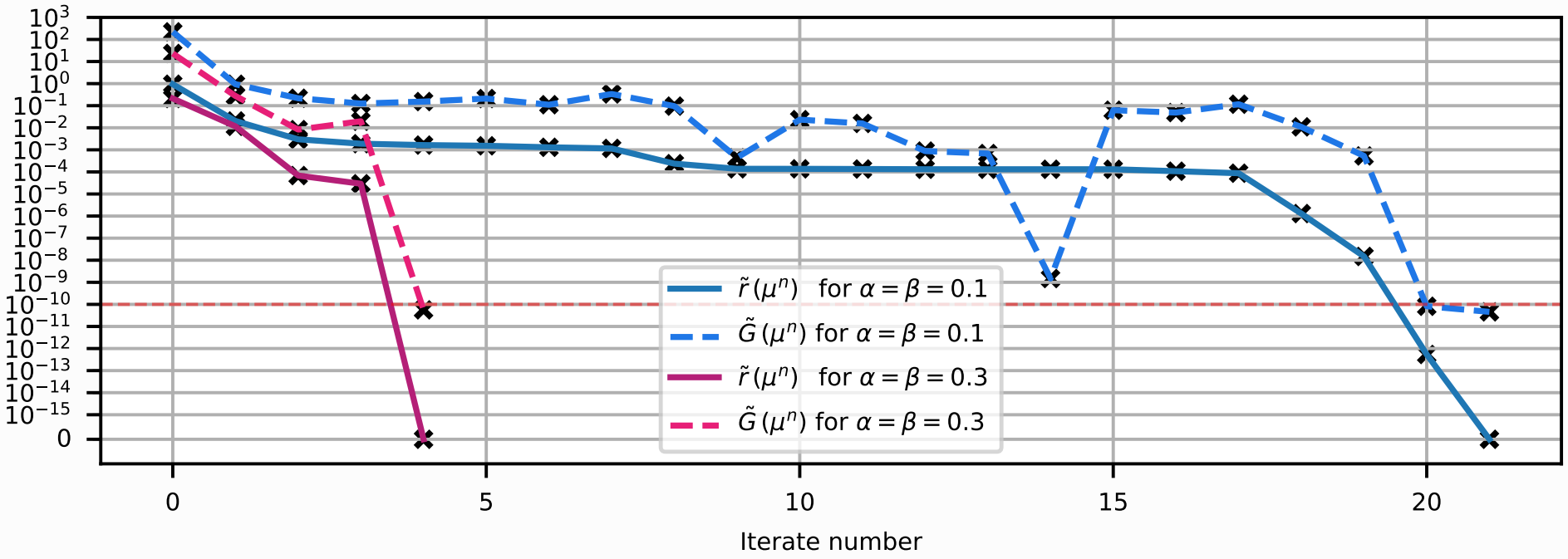
$\alpha = \beta = 0.3$



- **Low reg.** $\alpha, \beta = 0.1 \rightsquigarrow$ many low-energy artefacts around main trajectories
- **High reg.** $\alpha, \beta = 0.3 \rightsquigarrow$ improved reconstruction

Note! At each t_i the inverse problem $K^*_{t_i}\rho = f_{t_i}$ is heavily ill-posed: Indeed $K^*_{t_i}\delta_{\hat{x}} = K^*_{t_i}\delta_{\hat{x}+\lambda S_i^\perp}$ for $\lambda \in \mathbb{R}$, $S_i^\perp \perp L_i \rightsquigarrow$ Static methods cannot resolve location of $\hat{x}$

Convergence Plot



Note! Proven sublinear rate of convergence but empirical **linear rate**

As expected, higher regularization results in faster convergence

References

1. **Linear convergence of conditional gradient methods**
   - *Linear convergence of accelerated generalized conditional gradient methods*
     Preprint (2021), with K. Bredies, M. Carioni, D. Walter
2. **OT Regularization of Dynamic Inverse Problems**
   - *An optimal transport approach for solving dynamic inverse problems in spaces of measures*
     ESAIM: M2AN (2020), with K. Bredies
   - *A generalized conditional gradient method for dynamic inverse problems with optimal transport regularization*
     Found. of Comp. Math. (2021), with K. Bredies, M. Carioni, F. Romero
3. **Extremal Points of Transport Energies**
   - *On the extremal points of the ball of the Benamou-Brenier energy*
     Bull. London Math. Soc. (2021), with K. Bredies, M. Carioni, F. Romero
   - *A superposition principle for the inhomogeneous continuity equation with Hellinger-Kantorovich-regular coefficients*
     Communications in in PDE (2022), with K. Bredies, M. Carioni